

RAJALAKSHMI ENGINEERING COLLEGE
RAJALAKSHMI NAGAR, THANDALAM – 602 105



DEPARTMENT OF INFORMATION TECHNOLOGY

CS23621
Mobile Application Development
Laboratory

Laboratory Record Note Book

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**RAJALAKSHMI ENGINEERING
COLLEGE**
RAJALAKSHMI NAGAR, THANDALAM – 602 105

BONAFIDE CERTIFICATE

Name: MONIKA J **Academic Year:** 2025 – 2026

Semester: VI **Branch:** B.Tech - IT

Register No. 2116231001118

Certified that this is the Bonafide record of work done by the above student in the

*CS23621-Mobile-Application-Development-Laboratory Laboratory during the academic
year 2026 – 2027.*

Signature of Faculty in-charge

Submitted for the Practical Examination held on: 13-05-2026

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EX.NO : 1	Kotlin Basics
DATE:	

Aim

To study and implement basic Kotlin programming concepts such as variables, data types, operators, control statements, and null safety.

Algorithm

1. Open Android Studio.
2. Create a new Android project using Kotlin.
3. Design the layout using XML.
4. Write Kotlin code for variables, operators, loops, and null safety.
5. Run the application and verify the output.

Code

MainActivity.kt

```
package com.example.kotlinbasics
```

```
import android.os.Bundle
import android.widget.TextView
import androidx.appcompat.app.AppCompatActivity
```

```
class MainActivity : AppCompatActivity() {

    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        setContentView(R.layout.activity_main)

        val tvOutput = findViewById<TextView>(R.id.tvOutput)

        // Variables and Data Types
        val language = "Kotlin"
        var version = 1.9
        val isPopular = true

        // Operators
        val a = 10
        val b = 5

        val sum = a + b
        val difference = a - b

        // Control Statements
        val score = 85
    }
}
```

```
val grade = if(score >= 80) "B" else "C"
```

```
var evenNumbers = ""
```

```
for(i in 1..5) {  
    if(i % 2 == 0) {  
        evenNumbers += "$i "  
    }  
}
```

```
// While Loop
```

```
var count = 5
```

```
var countdown = ""
```

```
while(count > 0) {  
    countdown += "$count "  
    count--  
}
```

```
// Null Safety
```

```
val nullableString: String? = "Hello"
```

```
val safeCall = nullableString?.length ?: 0
```

```
val output = ""
```

```
    Variables and Data Types
```

```
    Language: $language
```

```
    Version: $version
```

```
    Popular: $isPopular
```

```
    Operators
```

```
    Sum: $sum
```

```
    Difference: $difference
```

```
    Control Statements
```

```
    Score: $score -> Grade: $grade
```

```
    Even Numbers: $evenNumbers
```

```
    Countdown: $countdown Blast off!
```

```
    Null Safety
```

```
    Nullable String: $nullableString
```

```
    Length: $safeCall
```

```
    """".trimIndent()
```

```
    tvOutput.text = output
```

```
    }  
}
```

activity_main.xml

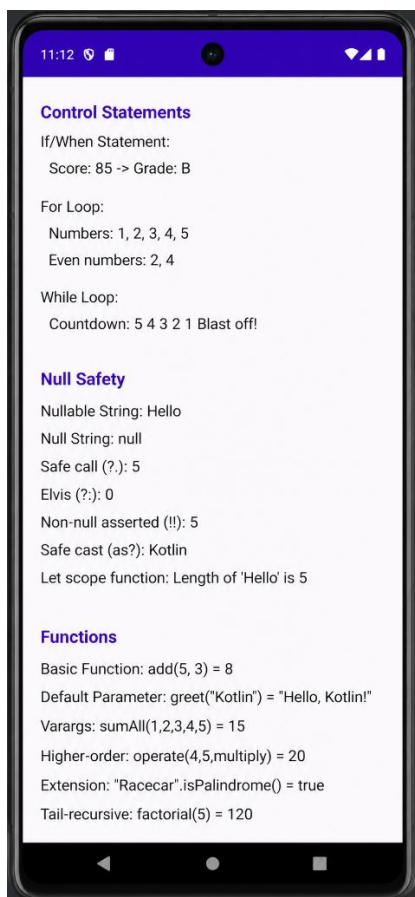
```
<?xml version="1.0" encoding="utf-8"?>

<ScrollView xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent">

    <TextView
        android:id="@+id/tvOutput"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:padding="20dp"
        android:textSize="18sp" />

</ScrollView>
```

Output



Result

Thus, the Kotlin basics program was successfully executed.

EX.NO : 2	Functions
DATE:	

Aim

To understand and implement different types of functions in Kotlin.

Algorithm

1. Open Android Studio.
2. Create a new Android project using Kotlin.
3. Design the layout using XML.
4. Write Kotlin functions with parameters and return types.
5. Call the functions inside MainActivity.kt.
6. Run the application and verify the output.

Code

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>

<ScrollView xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent">

    <TextView
        android:id="@+id/tvOutput"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:padding="20dp"
        android:textSize="18sp" />

</ScrollView>
```

MainActivity.kt

```
package com.example.functions

import android.os.Bundle
import android.widget.TextView
import androidx.appcompat.app.AppCompatActivity

class MainActivity : AppCompatActivity() {

    override fun onCreate(savedInstanceState: Bundle?) {
```

```

super.onCreate(savedInstanceState)
setContentView(R.layout.activity_main)

val tvOutput = findViewById<TextView>(R.id.tvOutput)

val addition = add(10, 5)
val subtraction = subtract(10, 5)
val multiplication = multiply(10, 5)
val factorialValue = factorial(5)

val output = ""
    Kotlin Functions

    add(10,5) = $addition

    subtract(10,5) = $subtraction

    multiply(10,5) = $multiplication

    factorial(5) = $factorialValue
"".trimIndent()

tvOutput.text = output
}

fun add(a: Int, b: Int): Int {
    return a + b
}

fun subtract(a: Int, b: Int): Int {
    return a - b
}

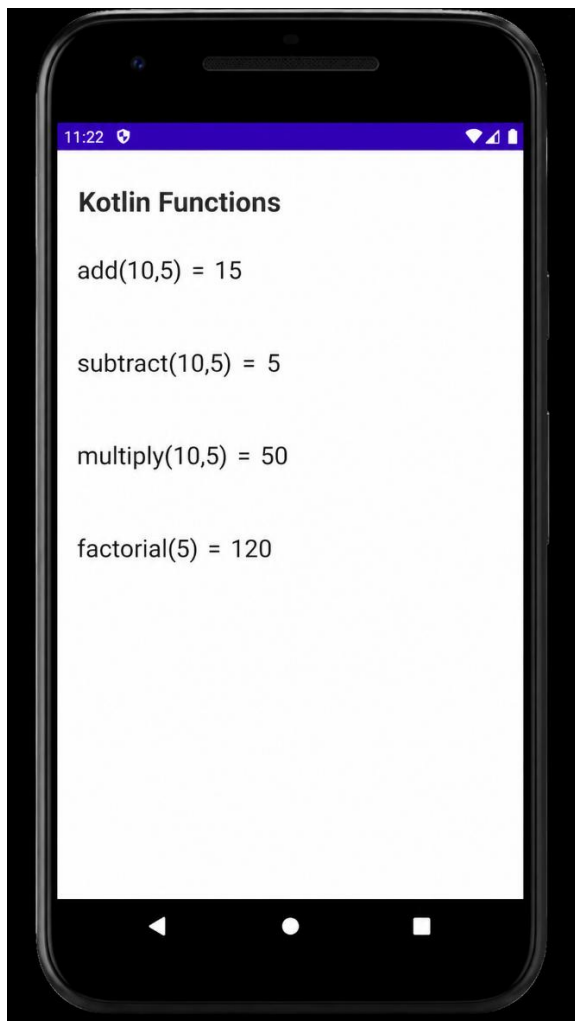
fun multiply(a: Int, b: Int): Int {
    return a * b
}

fun factorial(n: Int): Int {
    if (n == 1)
        return 1

    return n * factorial(n - 1)
}
}

```


Output



Result

Thus, the Kotlin functions program was successfully executed.

EX.NO : 3	Classes and Objects
DATE:	

Aim

To implement object-oriented programming concepts using classes and objects in Kotlin.

Algorithm

1. Open Android Studio.
2. Create a new Android project using Kotlin.
3. Design the layout using XML.
4. Create a class with properties and member functions.
5. Create objects and access class members.
6. Run the application and verify the output.

Code

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>

<ScrollView xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent">

    <TextView
        android:id="@+id/tvOutput"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:padding="20dp"
        android:textSize="18sp" />

</ScrollView>
```

MainActivity.kt

```
package com.example.classesobjects

import android.os.Bundle
import android.widget.TextView
import androidx.appcompat.app.AppCompatActivity

class Student(var name: String, var regNo: Int, var department: String) {

    fun displayDetails(): String {
```

```

        return ""
        Name: $name
        Register No: $regNo
        Department: $department
        """.trimIndent()
    }
}

class MainActivity : AppCompatActivity() {

    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        setContentView(R.layout.activity_main)

        val tvOutput = findViewById<TextView>(R.id.tvOutput)

        // Creating Object
        val student1 = Student(
            "Bhagya Lakshmi",
            2116231001025,
            "Information Technology"
        )

        val output = ""
        Classes and Objects

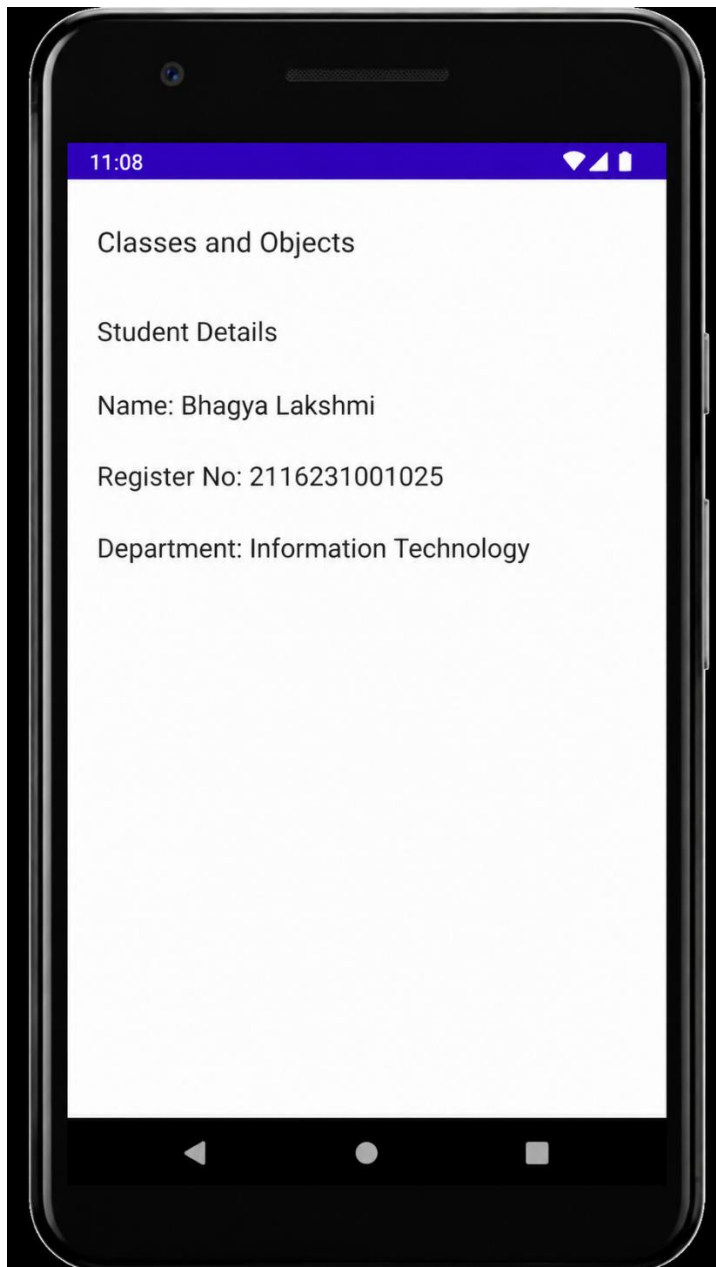
        Student Details

        ${student1.displayDetails()}
        """.trimIndent()

        tvOutput.text = output
    }
}

```

Output



Result

Thus, the classes and objects program was successfully executed.

EX.NO : 4	Build First Android App
DATE:	

Aim

To create and execute a basic Android application using Kotlin.

Algorithm

1. Start Android Studio IDE.
2. Create a new Android project using Kotlin.
3. Select “Empty Views Activity”.
4. Open activity_main.xml.
5. Create a LinearLayout.
6. Add a TextView component.
7. Open MainActivity.kt.
8. Initialize the TextView using findViewById().
9. Set text to the TextView.
10. Build and run the application.
11. Verify the output on emulator/device..

Code

activity_main.xml

```
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center"
    android:orientation="vertical">

    <TextView
        android:id="@+id/tvMessage"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:textSize="24sp" />

</LinearLayout>
```

MainActivity.kt

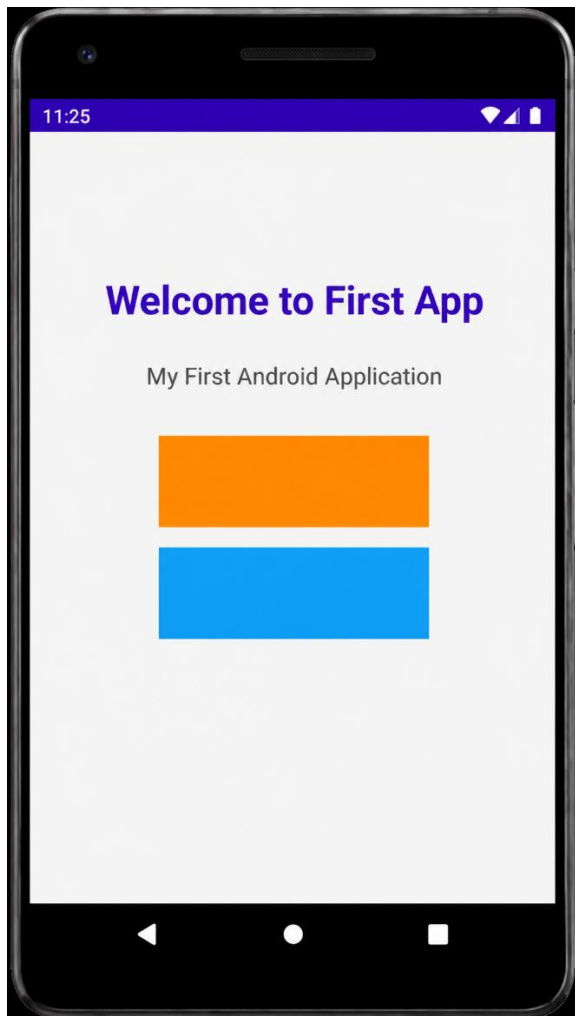
```
package com.example.firstapp
```

```
import android.os.Bundle
import android.widget.TextView
import androidx.appcompat.app.AppCompatActivity

class MainActivity : AppCompatActivity() {
    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        setContentView(R.layout.activity_main)

        val tvMessage = findViewById<TextView>(R.id.tvMessage)
        tvMessage.text = "Welcome to My First Android App"
    }
}
```

Output



Result

Thus, the first Android application was successfully executed.

EX.NO : 5	Layouts
DATE:	

Aim

To design user interfaces using Android layout managers.

Algorithm

1. Start Android Studio IDE.
2. Create a new Android project.
3. Open activity_main.xml.
4. Create a LinearLayout with vertical orientation.
5. Add TextView, EditText, and Button.
6. Add another TextView for displaying output.
7. Open MainActivity.kt.
8. Initialize UI components using findViewById().
9. Implement button click event using setOnClickListener.
10. Read text from EditText.
11. Display entered text in output TextView.
12. Build and run the application.
13. Verify the output.

Code

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>

<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:background="#F5F5F5">

    <TextView
        android:layout_width="match_parent"
        android:layout_height="60dp"
        android:gravity="center"
```

```
android:text="Login App"
android:textColor="#FFFFFF"
android:textSize="24sp"
android:textStyle="bold"
android:background="#6200EE" />
```

```
<LinearLayout
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:layout_margin="24dp"
    android:orientation="vertical"
    android:padding="20dp"
    android:background="#FFFFFF">
```

```
<TextView
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="User Login"
    android:textSize="24sp"
    android:textStyle="bold"
    android:textColor="#333333" />
```

```
<EditText
    android:id="@+id/etName"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:hint="Enter Username"
    android:layout_marginTop="20dp" />
```

```
<EditText
    android:id="@+id/etPassword"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:hint="Enter Password"
    android:inputType="textPassword"
    android:layout_marginTop="15dp" />
```

```
<Button
    android:id="@+id/btnSubmit"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="LOGIN"
    android:layout_marginTop="20dp"
    android:backgroundTint="#6200EE"
    android:textColor="#FFFFFF" />
```

```
<TextView
    android:id="@+id/tvResult"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:textSize="18sp"
    android:textColor="#000000"
```



```
        android:layout_marginTop="20dp" />
```

```
</LinearLayout>
```

```
</LinearLayout>
```

```
MainActivity.kt
```

```
package com.example.layouts
```

```
import android.os.Bundle
```

```
import android.widget.*
```

```
import androidx.appcompat.app.AppCompatActivity
```

```
class MainActivity : AppCompatActivity() {
```

```
    override fun onCreate(savedInstanceState: Bundle?) {
```

```
        super.onCreate(savedInstanceState)
```

```
        setContentView(R.layout.activity_main)
```

```
        val etName = findViewById<EditText>(R.id.etName)
```

```
        val etPassword = findViewById<EditText>(R.id.etPassword)
```

```
        val btnSubmit = findViewById<Button>(R.id.btnSubmit)
```

```
        val tvResult = findViewById<TextView>(R.id.tvResult)
```

```
        btnSubmit.setOnClickListener {
```

```
            val name = etName.text.toString()
```

```
            val password = etPassword.text.toString()
```

```
            if(name.isNotEmpty() && password.isNotEmpty()) {
```

```
                tvResult.text = "Hello, $name!"
```

```
            }
```

```
            else {
```

```
                tvResult.text = "Please enter username and password"
```

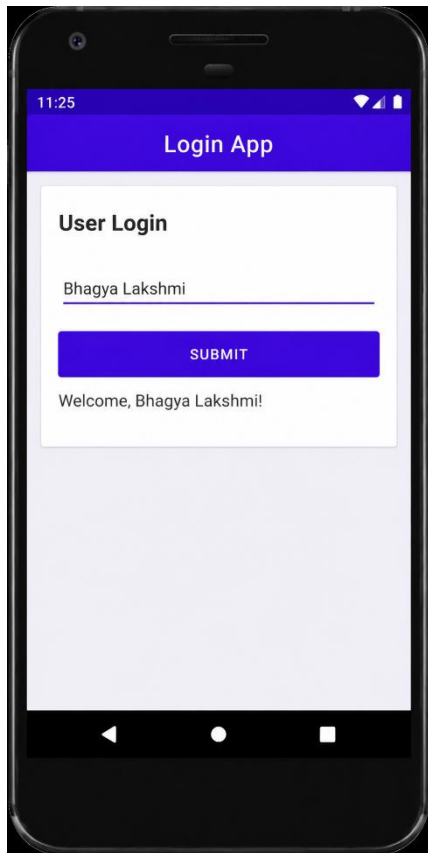
```
            }
```

```
        }
```

```
    }
```

```
}
```

Output



Result

Thus, Android layouts were successfully designed and executed.

EX.NO : 6	App Navigation	
DATE:		

Aim

To navigate from one activity to another activity using Intent.

Algorithm

1. Start Android Studio IDE.
2. Create a new Android project.
3. Create two activities:
4. MainActivity
5. SecondActivity
6. Design UI for both activities.
7. Add a button in MainActivity.
8. Open MainActivity.kt.
9. Initialize the button using findViewById().
10. Implement button click event.
11. Create Intent object.
12. Navigate to SecondActivity using startActivity().
13. Build and run the application.
14. Verify activity navigation.

Code

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>

<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center"
    android:orientation="vertical"
    android:background="#E8F5E9">

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Welcome to Second Activity"
```

```
        android:textSize="26sp"
        android:textStyle="bold"
        android:textColor="#2E7D32"/>
```

```
</LinearLayout>
```

activity_second.xml

```
<?xml version="1.0" encoding="utf-8"?>
```

```
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center"
    android:background="#F1F8E9"
    android:padding="20dp">
```

```
    <ImageView
        android:layout_width="120dp"
        android:layout_height="120dp"
        android:src="@android:drawable/checkbox_on_background"
        android:tint="#4CAF50" />
```

```
    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Congratulations!"
        android:textSize="32sp"
        android:textStyle="bold"
        android:textColor="#2E7D32"
        android:layout_marginTop="20dp" />
```

```
    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="You have successfully navigated\nto the Second Activity."
        android:textSize="20sp"
        android:textAlignment="center"
        android:textColor="#424242"
        android:layout_marginTop="15dp" />
```

```
</LinearLayout>
```

MainActivity.kt

```
package com.example.navigation
```

```
import android.content.Intent
import android.os.Bundle
import android.widget.Button
import androidx.appcompat.app.AppCompatActivity
```

```

class MainActivity : AppCompatActivity() {
    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        setContentView(R.layout.activity_main)

        val btnNext = findViewById<Button>(R.id.btnNext)

        btnNext.setOnClickListener {
            val intent = Intent(this, SecondActivity::class.java)
            startActivity(intent)
        }
    }
}

```

SecondActivity.kt

```

package com.example.navigation

import android.os.Bundle
import androidx.appcompat.app.AppCompatActivity

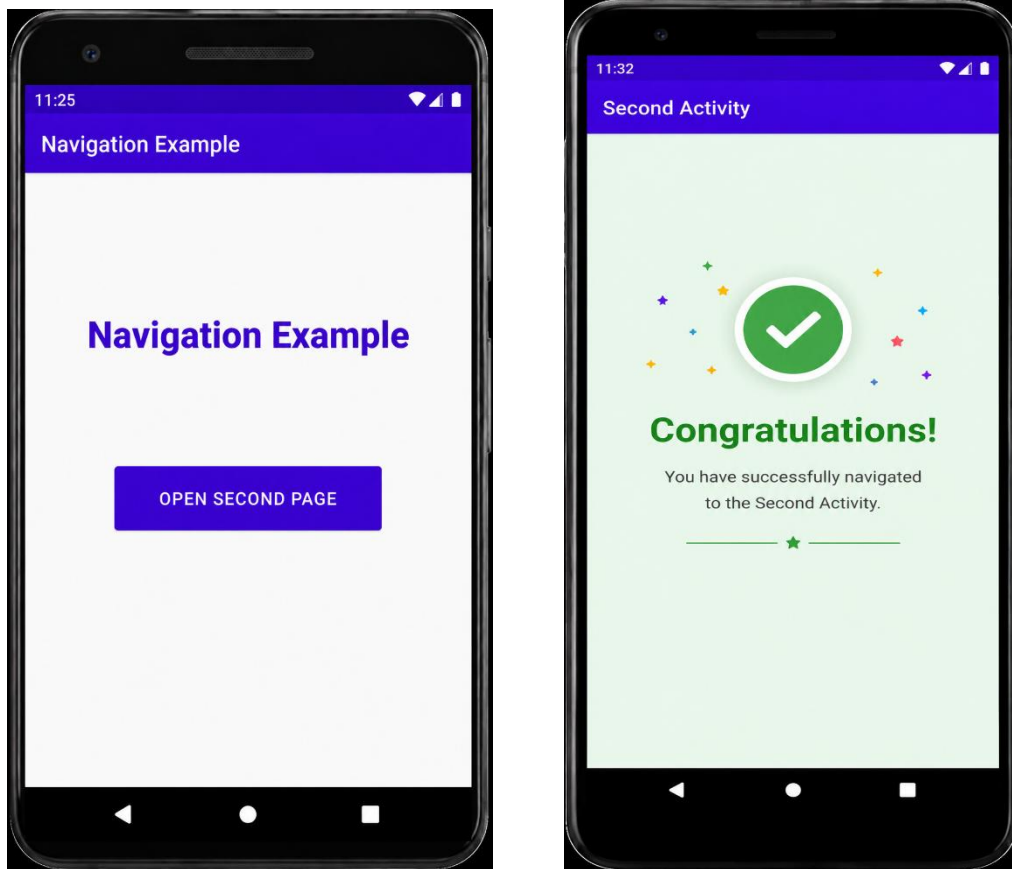
class SecondActivity : AppCompatActivity() {

    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)

        setContentView(R.layout.activity_second)
    }
}

```

Output



Result

Thus, app navigation using Intent was successfully implemented.

EX.NO : 7	Activity and Fragment Lifecycles
DATE:	

Aim

To study activity lifecycle methods in Android.

Algorithm

1. Start Android Studio IDE.
2. Create a new Android project.
3. Open activity_main.xml.
4. Add a TextView component.
5. Open MainActivity.kt.
6. Override lifecycle methods:
7. onCreate()
8. onStart()
9. onResume()
10. Display lifecycle method names in TextView.
11. Run the application.
12. Observe lifecycle execution order.
13. Verify the output.

Code

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>

<ScrollView xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:background="#F5F5F5">

    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:orientation="vertical">

        <!-- App Bar -->
```

```

<TextView
    android:layout_width="match_parent"
    android:layout_height="65dp"
    android:background="#6200EE"
    android:gravity="center"
    android:text="Student Details"
    android:textColor="#FFFFFF"
    android:textSize="26sp"
    android:textStyle="bold" />

<!-- Basic Details Fragment -->

<fragment
    android:id="@+id/fragmentBasic"
    android:name="org.rajalakshmi.androidfragments.StudentBasicDetails"
    android:layout_width="match_parent"
    android:layout_height="wrap_content" />

<!-- Mark Details Fragment -->

<fragment
    android:id="@+id/fragmentMark"
    android:name="org.rajalakshmi.androidfragments.StudentMarkDetails"
    android:layout_width="match_parent"
    android:layout_height="wrap_content" />

</LinearLayout>

```

```

</ScrollView>

```

fragment_student_basic_details.xml

```

<?xml version="1.0" encoding="utf-8"?>

<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:orientation="vertical"
    android:padding="20dp"
    android:layout_margin="15dp"
    android:background="#FFFFFF">

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Basic Details"
        android:textSize="24sp"
        android:textStyle="bold"
        android:textColor="#6200EE" />

    <EditText
        android:layout_width="match_parent"

```



```
    android:layout_height="wrap_content"
    android:hint="Register Number"
    android:layout_marginTop="20dp" />
```

```
<EditText
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:hint="Student Name"
    android:layout_marginTop="15dp" />
```

```
<EditText
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:hint="Department"
    android:layout_marginTop="15dp" />
```

```
</LinearLayout>
```

fragment_student_mark_details.xml

```
<?xml version="1.0" encoding="utf-8"?>
```

```
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:orientation="vertical"
    android:padding="20dp"
    android:layout_margin="15dp"
    android:background="#FFFFFF">
```

```
<TextView
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Mark Details"
    android:textSize="24sp"
    android:textStyle="bold"
    android:textColor="#4CAF50" />
```

```
<EditText
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:hint="S.S.L.C Mark"
    android:layout_marginTop="20dp" />
```

```
<EditText
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:hint="H.Sc Mark"
    android:layout_marginTop="15dp" />
```

```
<EditText
    android:layout_width="match_parent"
```

```
        android:layout_height="wrap_content"
        android:hint="U.G C.G.P.A"
        android:layout_marginTop="15dp" />
```

```
</LinearLayout>
```

MainActivity.kt

```
package org.rajalakshmi.androidfragments

import androidx.appcompat.app.AppCompatActivity
import android.os.Bundle

class MainActivity : AppCompatActivity() {

    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)

        setContentView(R.layout.activity_main)
    }
}
```

StudentBasicDetails.kt

```
package org.rajalakshmi.androidfragments

import androidx.fragment.app.Fragment

class StudentBasicDetails : Fragment(
    R.layout.fragment_student_basic_details
)
```

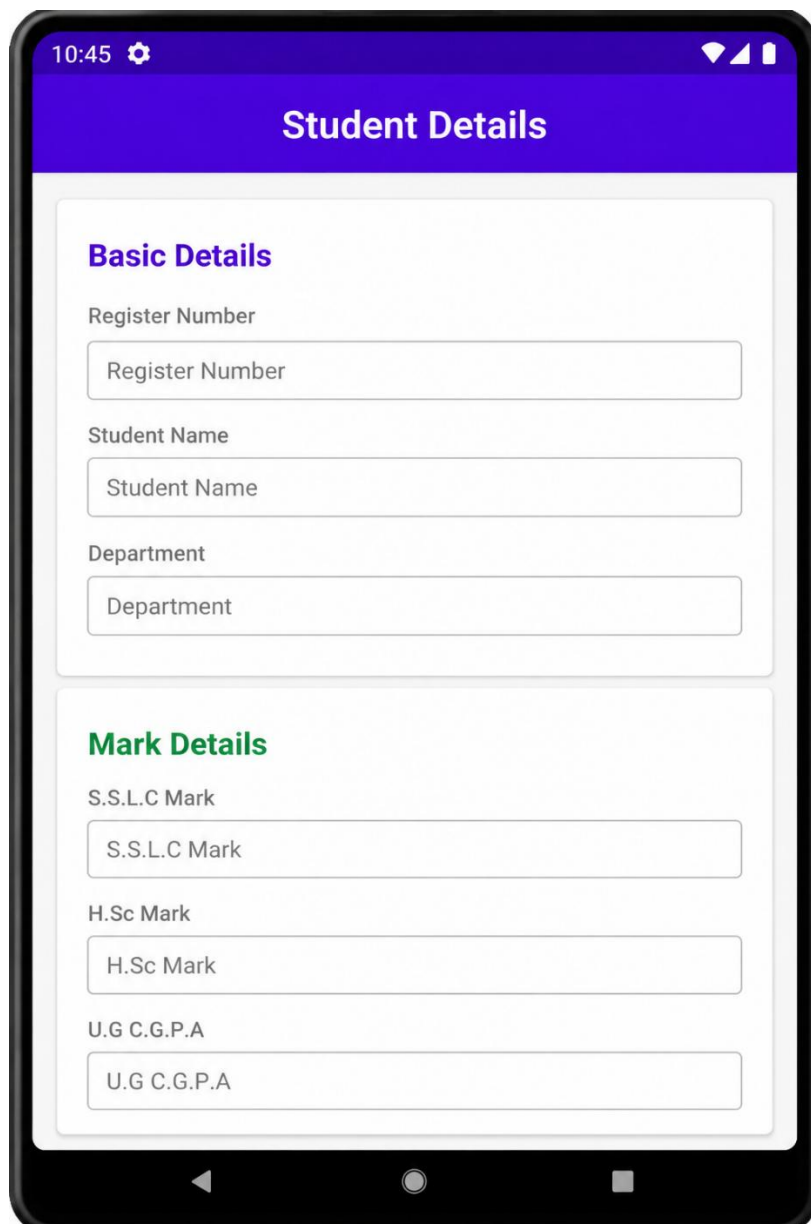
StudentMarkDetails.kt

```
package org.rajalakshmi.androidfragments

import androidx.fragment.app.Fragment

class StudentMarkDetails : Fragment(
    R.layout.fragment_student_mark_details
)
```

Output



The screenshot displays a mobile application interface with a purple header bar containing the title "Student Details". The app is running on a device with a status bar at the top showing the time as 10:45 and various icons. The main content area is divided into two sections: "Basic Details" and "Mark Details".

Basic Details

Register Number

Student Name

Department

Mark Details

S.S.L.C Mark

H.Sc Mark

U.G C.G.P.A

Result

Thus, activity lifecycle methods were successfully executed.

EX.NO : 8		App Architecture (UI Layer)
DATE:		

Aim

To implement UI Layer architecture using ViewModel and LiveData in Android.

Algorithm

1. Start Android Studio IDE.
2. Create a new Android project.
3. Create XML layout file.
4. Add UI components like TextView and Button.
5. Create ViewModel class.
6. Declare LiveData variable inside ViewModel.
7. Observe LiveData inside Activity.
8. Update UI using observed data.
9. Run the application.
10. Verify UI layer functionality.

Code

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>

<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:background="#F5F5F5">

    <TextView
        android:layout_width="match_parent"
        android:layout_height="65dp"
        android:gravity="center"
        android:text="UI Layer Architecture"
        android:textColor="#FFFFFF"
        android:textSize="24sp"
        android:textStyle="bold"
        android:background="#6200EE" />

    <LinearLayout
```

```
android:layout_width="match_parent"
android:layout_height="wrap_content"
android:layout_margin="24dp"
android:orientation="vertical"
android:padding="20dp"
android:background="#FFFFFF">
```

```
<TextView
    android:id="@+id/tvHeading"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Student Profile"
    android:textSize="24sp"
    android:textStyle="bold"
    android:textColor="#333333" />
```

```
<TextView
    android:id="@+id/tvName"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:textSize="20sp"
    android:layout_marginTop="20dp" />
```

```
<TextView
    android:id="@+id/tvDepartment"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:textSize="20sp"
    android:layout_marginTop="12dp" />
```

```
<TextView
    android:id="@+id/tvStatus"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:textSize="20sp"
    android:textColor="#2E7D32"
    android:textStyle="bold"
    android:layout_marginTop="12dp" />
```

```
</LinearLayout>
```

```
</LinearLayout>
```

MainActivity.kt

```
package com.example.uilayer
```

```
import android.os.Bundle
import android.widget.TextView
import androidx.appcompat.app.AppCompatActivity
import androidx.lifecycle.MutableLiveData
import androidx.lifecycle.ViewModel
```

```

import androidx.lifecycle.ViewModelProvider

class MainViewModel : ViewModel() {

    val name = MutableLiveData<String>()
    val department = MutableLiveData<String>()
    val status = MutableLiveData<String>()

    init {
        name.value = "Name: Bhagya Lakshmi"
        department.value = "Department: Information Technology"
        status.value = "Status: Data loaded from ViewModel"
    }
}

class MainActivity : AppCompatActivity() {

    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        setContentView(R.layout.activity_main)

        val tvName = findViewById<TextView>(R.id.tvName)
        val tvDepartment = findViewById<TextView>(R.id.tvDepartment)
        val tvStatus = findViewById<TextView>(R.id.tvStatus)

        val viewModel = ViewModelProvider(this)[MainViewModel::class.java]

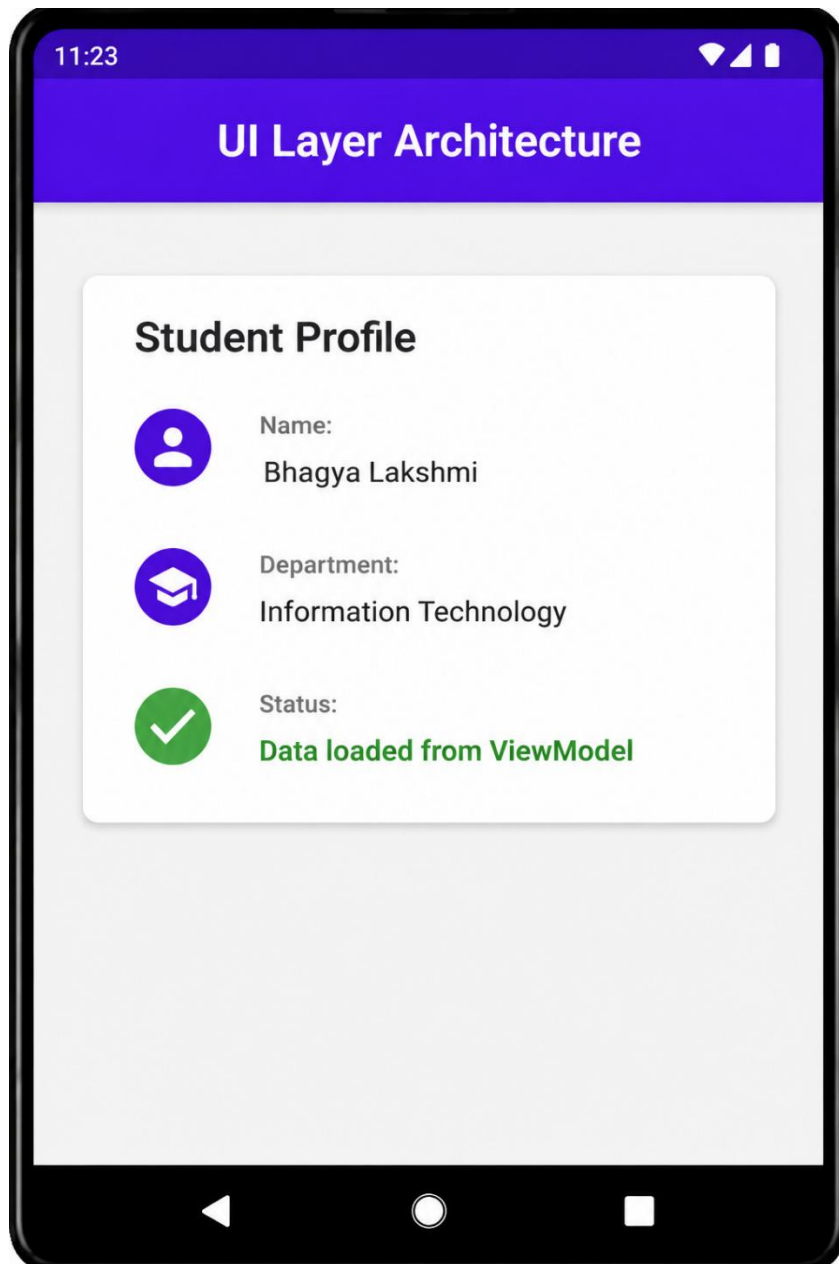
        viewModel.name.observe(this) {
            tvName.text = it
        }

        viewModel.department.observe(this) {
            tvDepartment.text = it
        }

        viewModel.status.observe(this) {
            tvStatus.text = it
        }
    }
}

```

Output



Result

Thus, the UI Layer architecture was successfully implemented using ViewModel and LiveData.

EX.NO : 9	App Architecture (Persistence)	
DATE:		

Aim

To implement data persistence using SQLite Database in Android.

Algorithm

1. Start Android Studio IDE.
2. Create a new Android project.
3. Add Room database dependencies.
4. Create Entity class.
5. Create DAO interface.
6. Create Room database class.
7. Insert data into database.
8. Retrieve stored data
9. Display data in UI.
10. Run the application.
11. Verify persistence functionality.

Code

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>

<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:background="#F5F5F5">

    <TextView
        android:layout_width="match_parent"
        android:layout_height="65dp"
        android:background="#6200EE"
        android:gravity="center"
        android:text="Student Database"
        android:textColor="#FFFFFF"
        android:textSize="24sp"
        android:textStyle="bold" />
```



```

<LinearLayout
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:layout_margin="24dp"
    android:orientation="vertical"
    android:padding="20dp"
    android:background="#FFFFFF">

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Save Student Record"
        android:textSize="24sp"
        android:textStyle="bold"
        android:textColor="#333333" />

    <EditText
        android:id="@+id/etName"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginTop="20dp"
        android:hint="Enter Student Name" />

    <Button
        android:id="@+id/btnSave"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="SAVE DATA"
        android:textColor="#FFFFFF"
        android:backgroundTint="#6200EE"
        android:layout_marginTop="20dp" />

    <TextView
        android:id="@+id/tvStatus"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:textSize="18sp"
        android:textColor="#2E7D32"
        android:textStyle="bold"
        android:layout_marginTop="20dp" />

```

```

</LinearLayout>

```

```

</LinearLayout>

```

MainActivity.kt

```

package com.example.persistence

```

```

import android.database.sqlite.SQLiteDatabase
import android.os.Bundle

```

```

import android.widget.*
import androidx.appcompat.app.AppCompatActivity

class MainActivity : AppCompatActivity() {

    lateinit var database: SQLiteDatabase

    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)

        setContentView(R.layout.activity_main)

        val etName = findViewById<EditText>(R.id.etName)
        val btnSave = findViewById<Button>(R.id.btnSave)
        val tvStatus = findViewById<TextView>(R.id.tvStatus)

        database = openOrCreateDatabase(
            "StudentDB",
            MODE_PRIVATE,
            null
        )

        database.execSQL(
            "CREATE TABLE IF NOT EXISTS Student(name VARCHAR)"
        )

        btnSave.setOnClickListener {

            val studentName = etName.text.toString()

            if(studentName.isNotEmpty()) {

                database.execSQL(
                    "INSERT INTO Student VALUES('$studentName')"
                )

                tvStatus.text = "Record Saved Successfully"

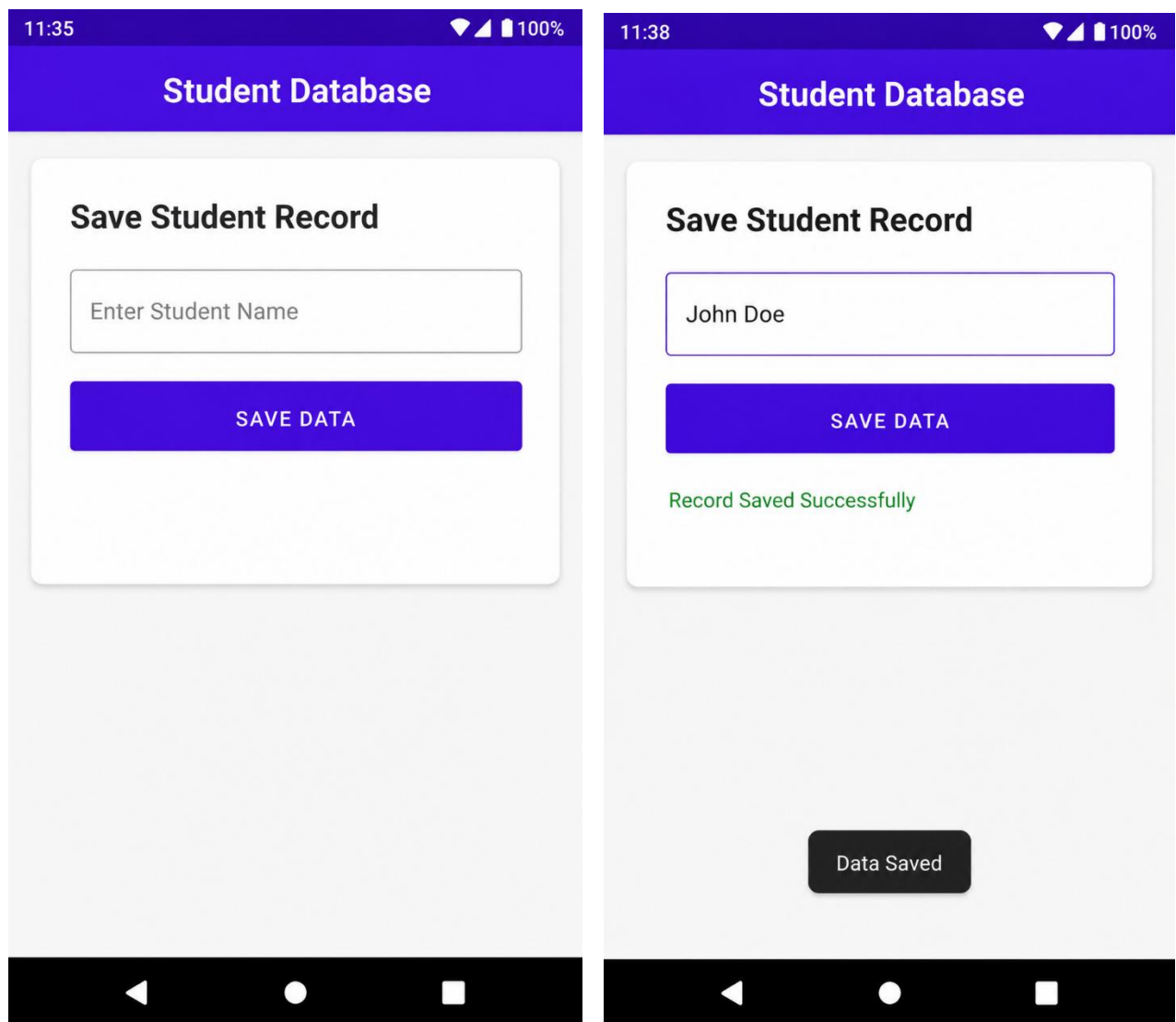
                Toast.makeText(
                    this,
                    "Data Saved",
                    Toast.LENGTH_SHORT
                ).show()

            } else {

                tvStatus.text = "Please enter student name"
            }
        }
    }
}

```

Output



Result

Thus, data persistence using SQLite database was successfully implemented.

EX.NO : 10		Advanced RecyclerView Use Cases
DATE:		

Aim

To implement RecyclerView for displaying list items in Android.

Algorithm

1. Start Android Studio IDE.
2. Create a new Android project.
3. Add RecyclerView dependency.
4. Open XML layout file.
5. Add RecyclerView component.
6. Create item layout XML.
7. Create Adapter class.
8. Create ViewHolder class.
9. Prepare data list.
10. Set LayoutManager for RecyclerView.
11. Attach adapter to RecyclerView.
12. Run the application.
13. Verify list display.

Code

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>

<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:background="#F5F5F5">

    <TextView
        android:layout_width="match_parent"
        android:layout_height="65dp"
        android:gravity="center"
        android:text="RecyclerView Example"
        android:textColor="#FFFFFF"
```

```
        android:textSize="24sp"
        android:textStyle="bold"
        android:background="#6200EE"/>
```

```
<androidx.recyclerview.widget.RecyclerView
    android:id="@+id/recyclerView"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:padding="10dp"/>
```

</LinearLayout>

item_layout.xml

```
<?xml version="1.0" encoding="utf-8"?>
```

```
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:orientation="vertical"
    android:padding="16dp"
    android:layout_margin="10dp"
    android:background="#FFFFFF">
```

```
<TextView
    android:id="@+id/tvItem"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:textSize="22sp"
    android:textStyle="bold"
    android:textColor="#6200EE" />
```

```
<TextView
    android:id="@+id/tvDescription"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Course Module"
    android:textSize="16sp"
    android:layout_marginTop="6dp" />
```

</LinearLayout>

MyAdapter.kt

```
package com.example.recyclerview
```

```
import android.view.LayoutInflater
import android.view.View
import android.view.ViewGroup
import android.widget.TextView
import androidx.recyclerview.widget.RecyclerView
```

```

class MyAdapter(private val list: ArrayList<String>) :
    RecyclerView.Adapter<MyAdapter.ViewHolder>() {

    class ViewHolder(itemView: View) : RecyclerView.ViewHolder(itemView) {

        val tvItem: TextView =
            itemView.findViewById(R.id.tvItem)

        val tvDescription: TextView =
            itemView.findViewById(R.id.tvDescription)
    }

    override fun onCreateViewHolder(
        parent: ViewGroup,
        viewType: Int
    ): ViewHolder {

        val view = LayoutInflater.from(parent.context)
            .inflate(R.layout.item_layout, parent, false)

        return ViewHolder(view)
    }

    override fun onBindViewHolder(
        holder: ViewHolder,
        position: Int
    ) {

        holder.tvItem.text = list[position]

        holder.tvDescription.text =
            "Learning Module ${position + 1}"
    }

    override fun getItemCount(): Int {
        return list.size
    }
}

```

MainActivity.kt

```

package com.example.recyclerview

import android.os.Bundle
import androidx.appcompat.app.AppCompatActivity
import androidx.recyclerview.widget.LinearLayoutManager
import androidx.recyclerview.widget.RecyclerView

class MainActivity : AppCompatActivity() {

    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
    }
}

```

```
setContentView(R.layout.activity_main)

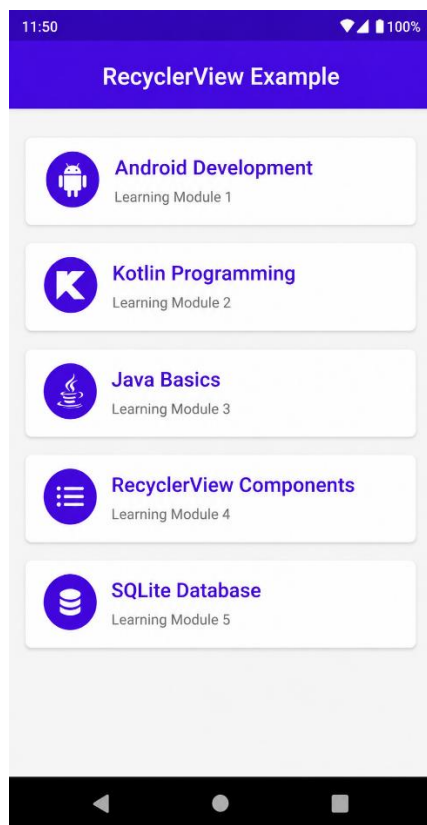
val recyclerView =
    findViewById<RecyclerView>(R.id.recyclerView)

val list = arrayListOf(
    "Android Development",
    "Kotlin Programming",
    "Java Basics",
    "RecyclerView Components",
    "SQLite Database"
)

recyclerView.layoutManager =
    LinearLayoutManager(this)

recyclerView.adapter =
    MyAdapter(list)
}
```

Output



Result

Thus, RecyclerView was successfully implemented for displaying list items.

EX.NO : 11		Connect to the Internet
DATE:		

Aim

To connect Android application to the Internet..

Algorithm

1. Start Android Studio IDE.
2. Create a new Android project.
3. Open AndroidManifest.xml.
4. Add Internet permission.
5. Create UI components.
6. Use URL connection or API request.
7. Fetch data from Internet.
8. Display received data in UI.
9. Build and run the application.
10. Verify Internet connection.

Code

AndroidManifest.xml

```
<uses-permission android:name="android.permission.INTERNET"/>
```

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
```

```
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:background="#F5F5F5">
```

```
<!-- App Bar -->
```

```
<TextView
    android:layout_width="match_parent"
```



```
        android:layout_height="65dp"
        android:background="#6200EE"
        android:gravity="center"
        android:text="Internet Connection"
        android:textColor="#FFFFFF"
        android:textSize="24sp"
        android:textStyle="bold"/>
```

```
<!-- Card -->
```

```
<LinearLayout
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:layout_margin="24dp"
    android:orientation="vertical"
    android:padding="20dp"
    android:background="#FFFFFF">
```

```
    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Network Status"
        android:textSize="24sp"
        android:textStyle="bold"
        android:textColor="#333333"/>
```

```
    <TextView
        android:id="@+id/tvData"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:textSize="20sp"
        android:textColor="#2E7D32"
        android:layout_marginTop="25dp"/>
```

```
</LinearLayout>
```

```
</LinearLayout>
```

```
MainActivity.kt
```

```
package com.example.internet
```

```
import android.os.Bundle
```

```
import android.widget.TextView
```

```
import androidx.appcompat.app.AppCompatActivity
```

```
class MainActivity : AppCompatActivity() {
```

```
    override fun onCreate(savedInstanceState: Bundle?) {
```

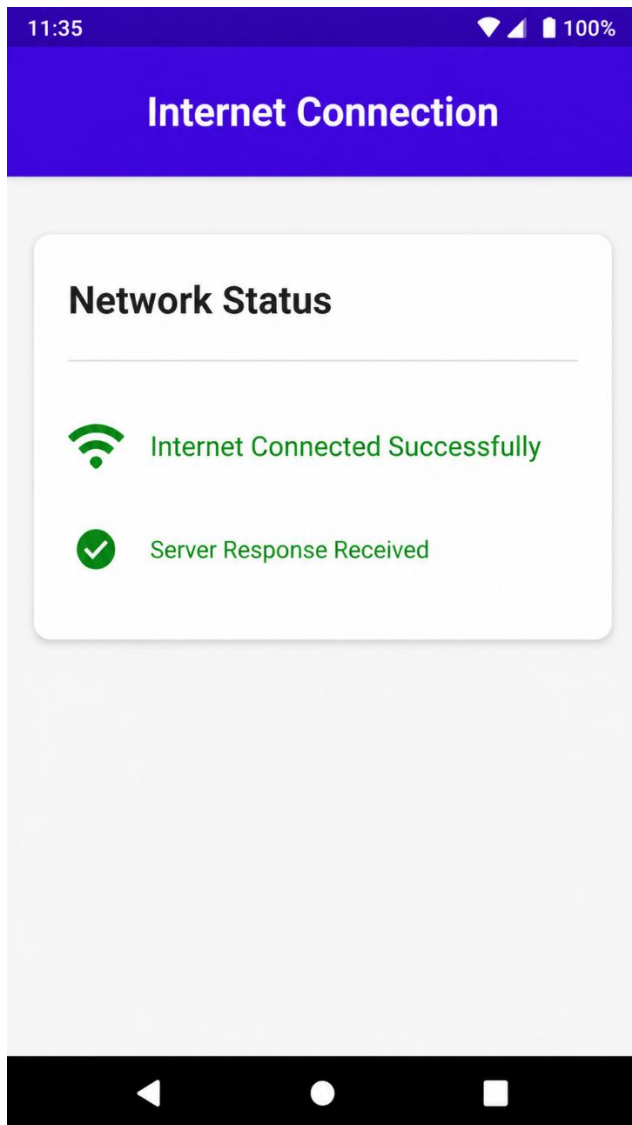
```
        super.onCreate(savedInstanceState)
```

```
setContentView(R.layout.activity_main)

val tvData =
    findViewById<TextView>(R.id.tvData)

tvData.text =
    "Internet Connected Successfully\n\nServer Response Received"
}
```

Output



Result

Thus, the Android application was successfully connected to the Internet.

EX.NO : 12	Send Email
DATE:	

Aim

To send email using Intent in Android..

Algorithm

1. Start Android Studio IDE.
2. Create a new Android project.
3. Design UI with button.
4. Open MainActivity.kt.
5. Initialize button using findViewById().
6. Implement button click event.
7. Create email Intent.
8. Set email address, subject, and message.
9. Launch email application using Intent.
10. Run the application.
11. Verify email sending functionality..

Code

activity_main.xml

```
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center">

    <Button
        android:id="@+id/btnEmail"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Send Email"/>
</LinearLayout>
```

MainActivity.kt

```
package com.example.sendemail
```

```
import android.content.Intent
import android.os.Bundle
import android.widget.Button
```

```
import androidx.appcompat.app.AppCompatActivity

class MainActivity : AppCompatActivity() {

    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        setContentView(R.layout.activity_main)

        val btnEmail = findViewById<Button>(R.id.btnEmail)

        btnEmail.setOnClickListener {

            val intent = Intent(Intent.ACTION_SEND)

            intent.type = "message/rfc822"

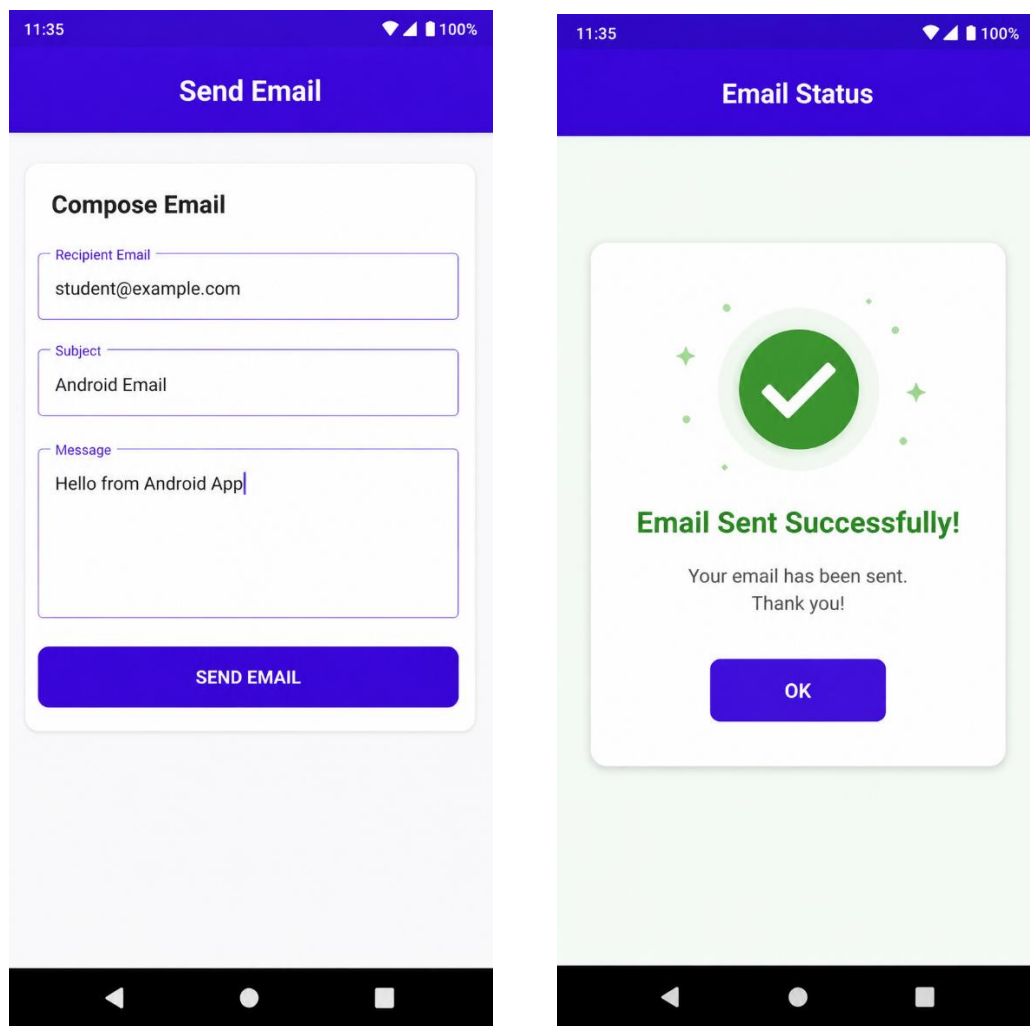
            intent.putExtra(
                Intent.EXTRA_EMAIL,
                arrayOf("test@gmail.com")
            )

            intent.putExtra(
                Intent.EXTRA_SUBJECT,
                "Android Email"
            )

            intent.putExtra(
                Intent.EXTRA_TEXT,
                "Hello from Android App"
            )

            startActivity(intent)
        }
    }
}
```

Output



Result

Thus, email sending using Intent was successfully implemented.

EX.NO : 13	Repository Pattern and WorkManager
DATE:	

Aim

To implement Repository Pattern and WorkManager in Android.

Algorithm

1. Start Android Studio IDE.
2. Create a new Android project.
3. Add WorkManager dependency.
4. Create Worker class.
5. Override doWork() method.
6. Create Repository class.
7. Create WorkRequest object.
8. Enqueue background task using WorkManager.
9. Execute background process.
10. Run the application.
11. Verify WorkManager execution.

Code

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>

<ScrollView xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:background="#F5F5F5">

    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:orientation="vertical"
        android:padding="16dp">

        <TextView
            android:layout_width="match_parent"
            android:layout_height="60dp"
            android:gravity="center"
            android:text="Repository & WorkManager"
            android:textColor="#FFFFFF"
```

```
        android:textSize="22sp"
        android:textStyle="bold"
        android:background="#3F51B5" />
```

```
<LinearLayout
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:orientation="horizontal"
    android:gravity="center"
    android:layout_marginTop="16dp">
```

```
    <Button
        android:id="@+id/btnRefresh"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Refresh Users" />
```

```
    <Button
        android:id="@+id/btnSync"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Start Sync"
        android:layout_marginLeft="12dp" />
```

```
</LinearLayout>
```

```
<TextView
    android:id="@+id/tvStatus"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="Status: Waiting for background sync"
    android:textSize="17sp"
    android:textColor="#2E7D32"
    android:layout_marginTop="16dp"
    android:padding="12dp"
    android:background="#E8F5E9" />
```

```
<TextView
    android:id="@+id/tvUsers"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:textSize="17sp"
    android:layout_marginTop="16dp"
    android:lineSpacingExtra="8dp" />
```

```
</LinearLayout>
```

```
</ScrollView>
```

MainActivity.kt

```
package com.example.workmanager
```

```

import android.content.Context
import android.os.Bundle
import android.widget.Button
import android.widget.TextView
import androidx.appcompat.app.AppCompatActivity
import androidx.work.OneTimeWorkRequestBuilder
import androidx.work.WorkManager
import androidx.work.Worker
import androidx.work.WorkerParameters

data class User(
    val name: String,
    val email: String,
    val phone: String,
    val synced: Boolean
)

class UserRepository {

    fun getUsers(): List<User> {
        return listOf(
            User("John Doe", "john@example.com", "555-1234", true),
            User("Jane Smith", "jane@example.com", "555-5678", false),
            User("Bob Johnson", "bob@example.com", "555-9012", true),
            User("New User", "new@example.com", "123-456-7890", false)
        )
    }
}

class SyncWorker(
    context: Context,
    params: WorkerParameters
) : Worker(context, params) {

    override fun doWork(): Result {
        println("Background sync task running")
        return Result.success()
    }
}

class MainActivity : AppCompatActivity() {

    private val repository = UserRepository()

    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)

        setContentView(R.layout.activity_main)

        val btnRefresh = findViewById<Button>(R.id.btnRefresh)
        val btnSync = findViewById<Button>(R.id.btnSync)
        val tvStatus = findViewById<TextView>(R.id.tvStatus)
    }
}

```



```

val tvUsers = findViewById<TextView>(R.id.tvUsers)

btnRefresh.setOnClickListener {
    val users = repository getUsers()

    var output = ""

    for (user in users) {
        output += ""
        ${user.name}
        ${user.email}
        ${user.phone}
        Synced: ${if (user.synced) "Yes" else "No"}

        """".trimIndent() + "\n\n"
    }

    tvUsers.text = output
    tvStatus.text = "Users refreshed from repository!"
}

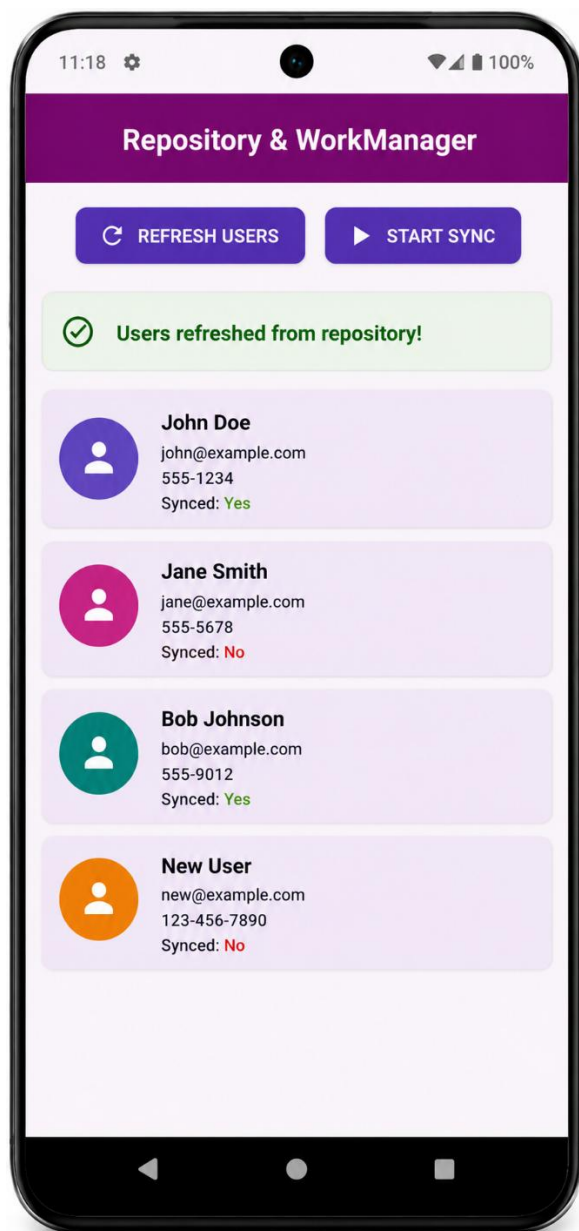
btnSync.setOnClickListener {
    val workRequest =
        OneTimeWorkRequestBuilder<SyncWorker>().build()

    WorkManager
        .getInstance(this)
        .enqueue(workRequest)

    tvStatus.text = "WorkManager background sync started!"
}
}
}

```

Output



Result

Thus, Repository Pattern and WorkManager were successfully implemented.

EX.NO : 14		App UI Design
DATE:		

Aim

To design attractive Android user interface using XML layouts.

Algorithm

1. Start Android Studio IDE.
2. Create a new Android project.
3. Add WorkManager dependency.
4. Create Worker class.
5. Override doWork() method.
6. Create Repository class.
7. Create WorkRequest object.
8. Enqueue background task using WorkManager.
9. Execute background process.
10. Run the application.
11. Verify WorkManager execution.

Code

1. Create New Project

- Open Android Studio
- Click New Project
- Select Empty Activity
- Language: Kotlin
- Minimum SDK: 24

2. Add Dependencies

app/build.gradle.kts

```

plugins{
    id("com.android.application")
    id("org.jetbrains.kotlin.android")
}

android{
    namespace = "com.example.attendancemanager"
    compileSdk = 34

    defaultConfig {
        applicationId = "com.example.attendancemanager"
        minSdk = 24
        targetSdk = 34
        versionCode = 1
        versionName = "1.0"
    }

    buildFeatures {
        compose = true
    }

    composeOptions {
        kotlinCompilerExtensionVersion = "1.5.1"
    }

    kotlinOptions {
        jvmTarget = "1.8"
    }
}

dependencies {

    implementation("androidx.core:core-ktx:1.12.0")
    implementation("androidx.lifecycle:lifecycle-runtime-ktx:2.6.2")
    implementation("androidx.activity:activity-compose:1.8.2")

    implementation(platform("androidx.compose:compose-bom:2023.10.01"))
    implementation("androidx.compose.ui:ui")
    implementation("androidx.compose.ui:ui-graphics")
    implementation("androidx.compose.ui:ui-tooling-preview")
    implementation("androidx.compose.material3:material3")

    implementation("androidx.navigation:navigation-compose:2.7.5")

    debugImplementation("androidx.compose.ui:ui-tooling")
}

```

3. AndroidManifest.xml

```
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
    android:versionCode="1"
    android:versionName="1.0">

    <application
        android:allowBackup="true"
        android:label="@string/app_name"
        android:supportRtl="true"
        android:theme="@style/Theme.AppCompat.Light.NoActionBar">

        <activity
            android:name=".MainActivity"
            android:exported="true">

            <intent-filter>
                <action android:name="android.intent.action.MAIN">
                <category android:name="android.intent.category.LAUNCHER">
            </intent-filter>

        </activity>
    </application>
</manifest>
```

4. Student Data Class

Student.kt

```
package com.example.attendancemanager

data class Student(
    val name: String
    var isPresent: Boolean = false
)
```

5. MainActivity.kt

```
package com.example.attendancemanager

import android.os.Bundle
import androidx.activity.ComponentActivity
import androidx.activity.compose.setContent
import androidx.compose.foundation.background
import androidx.compose.foundation.layout.*
import androidx.compose.foundation.lazy.LazyColumn
import androidx.compose.foundation.lazy.items
import androidx.compose.foundation.shape.RoundedCornerShape
import androidx.compose.material3.*
import androidx.compose.runtime.*
```

```

androidx.compose.ui.Alignment import androidx.compose.ui.Modifier import
androidx.compose.ui.graphics.Brush import
androidx.compose.ui.graphics.Color import
androidx.compose.ui.text.font.FontWeight import
androidx.compose.ui.unit.dp import androidx.compose.ui.unit.sp
class MainActivity : ComponentActivity() {

    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)

        setContent {
            AttendanceApp()
        }
    }

    @Composable
    var studentName by remember { mutableStateOf("") }

    val studentList = remember { mutableStateListOf<Student>() }

    Column( modifier = Modifier
.fillMaxSize()
.background(
    Brush.verticalGradient( listOf(Color(0xFFE3F2FD),
        Color.White)
    )
)
.padding(16.dp)
) {

    Text( text = "Attendance Manager",
        fontSize = 30.sp, fontWeight =
        FontWeight.Bold, color =
        Color(0xFF0D47A1)
    )

    Spacer(modifier = Modifier.height(16.dp))

    OutlinedTextField( value =
        studentName, onValueChange = {
studentName = it
    },
    label = {
Text("Enter Student Name")
    },
    modifier = Modifier.fillMaxWidth()
)

    Spacer(modifier = Modifier.height(10.dp))

```

```

Button( onClick = { if (studentName.isNotEmpty()) {
    studentList.add(Student(studentName))
    studentName = ""
}
},
modifier = Modifier.fillMaxWidth()
) {
Text("Add Student")
}

Spacer(modifier = Modifier.height(20.dp))

LazyColumn { items(studentList) { student ->

    Card( modifier = Modifier .fillMaxWidth()
        .padding(bottom = 12.dp), shape =
RoundedCornerShape(20.dp),
elevation = CardDefaults.cardElevation(8.dp)
) {

        Row( modifier = Modifier .fillMaxWidth()
            .padding(16.dp), verticalAlignment = Alignment.CenterVertically,
horizontalArrangement = Arrangement.SpaceBetween
) {

            Column {

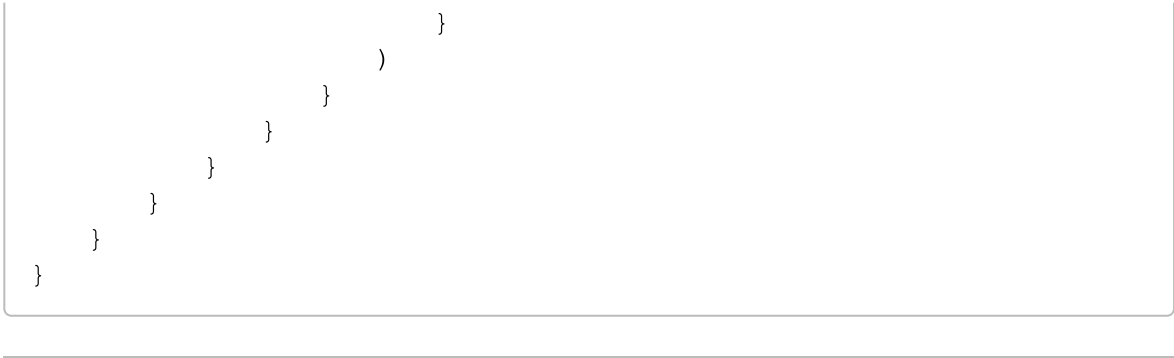
                Text( text = student.name, fontSize = 22.sp, fontWeight =
                    FontWeight.Bold
                )

                Spacer(modifier = Modifier.height(6.dp))

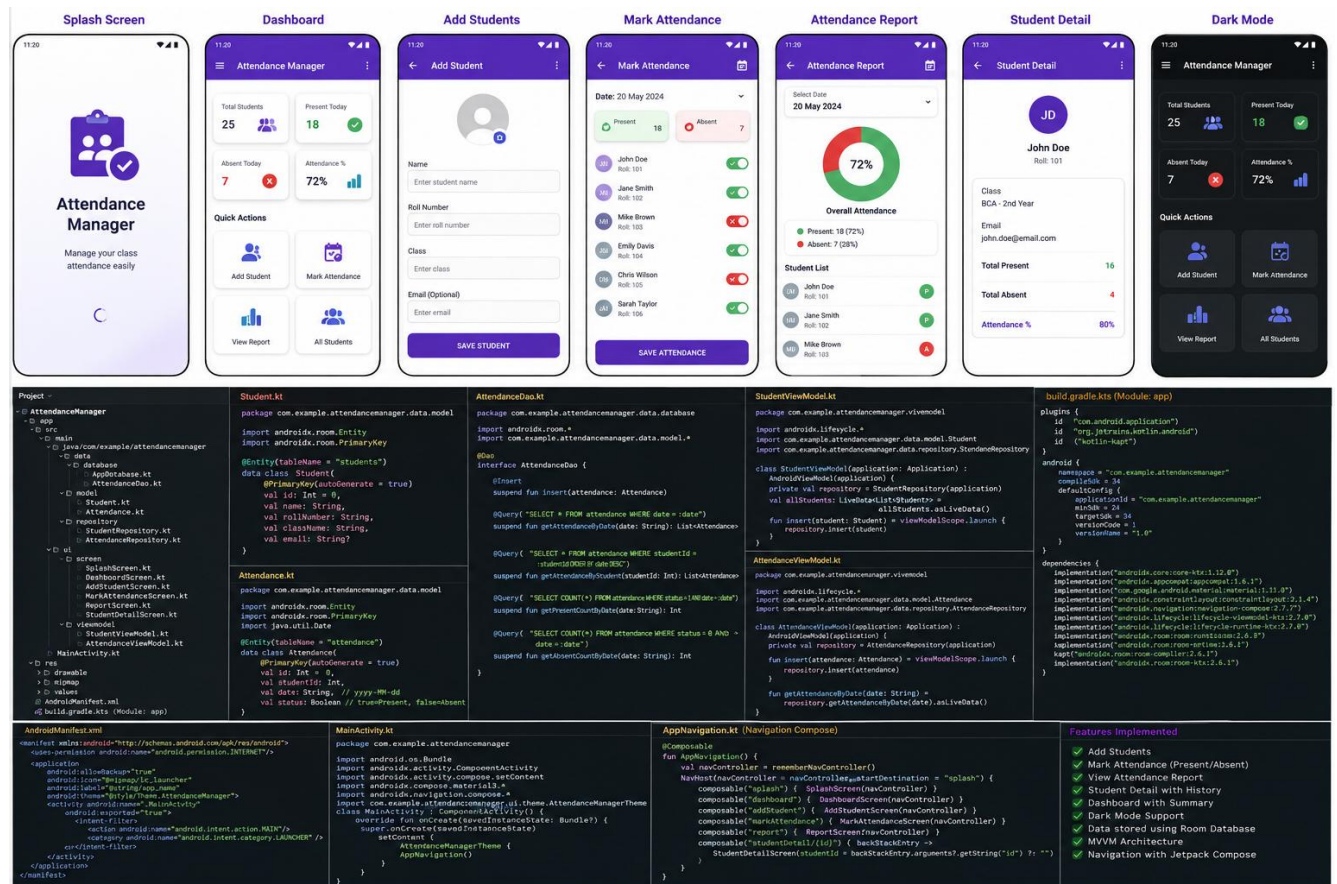
                Text( text = if (student.isPresent)
                    "Present" else
                    "Absent", color = if (student.isPresent)
                    Color(0xFF2E7D32) else
                        Color.Red
                )
            }

            Switch( checked = student.isPresent, onCheckedChange =
                { student.isPresent = it

```



Output



Result

Thus, attractive Android UI design was successfully implemented using XML layouts.